MicroVLA

Architecture Reference

~30M deployed · frozen YOLO-World-S + recursive world model + chrono planner

Pipeline

2 Hz vision → 30 Hz
JEPA dream loop →
plan [5×7]

TRM pattern

x / y / z recursion
weight-tied TinyNet
residual next_emb

Losses & tasks

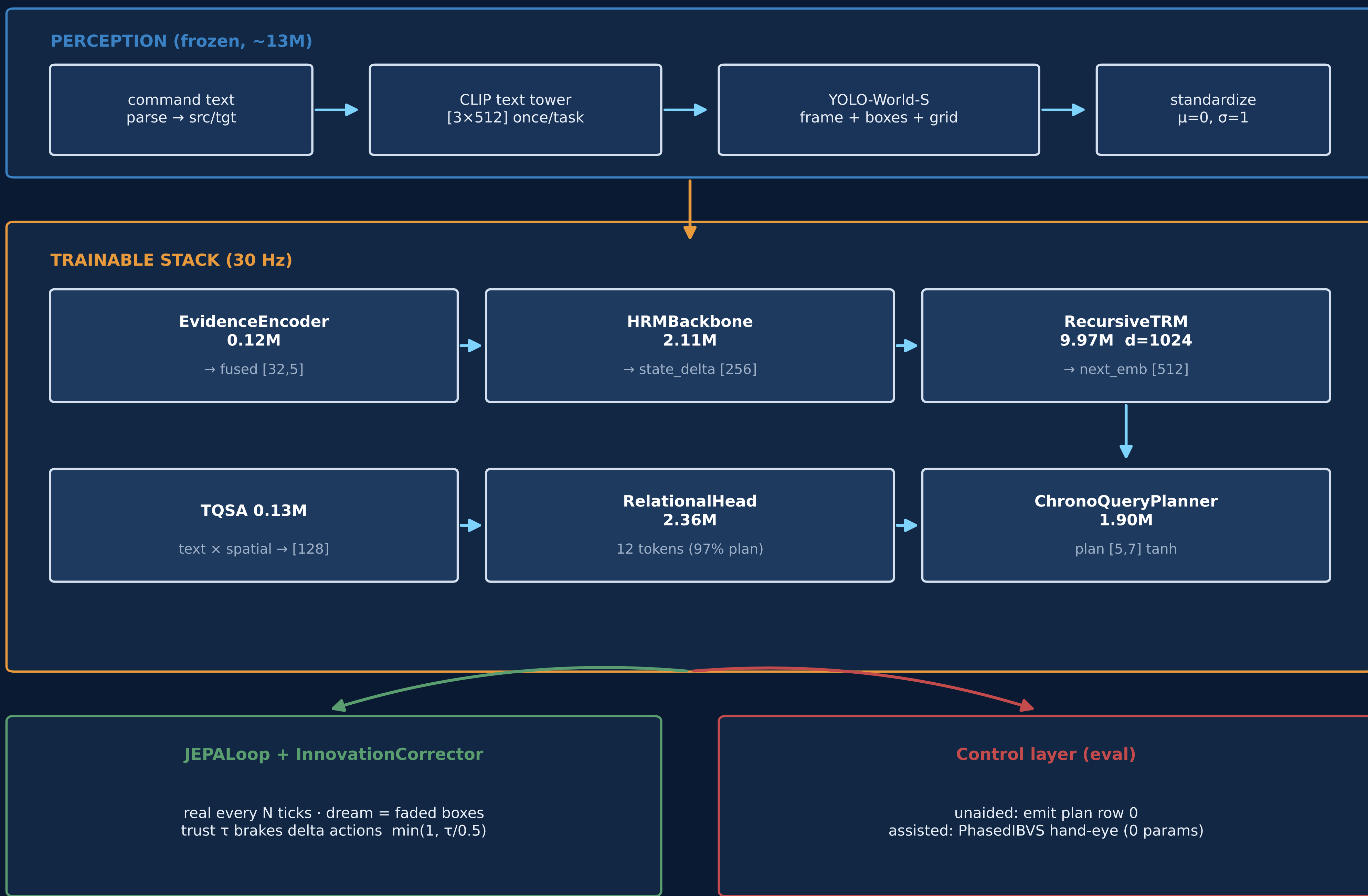
spec · BC · grip BCE
LIBERO object suite
aided vs unaided

Checkpoint of record: full_stageB_rec_fix.pt

Trainable ≈ 16.6M · Frozen detector ≈ 13M · Target: Pi 5 / 7 servos

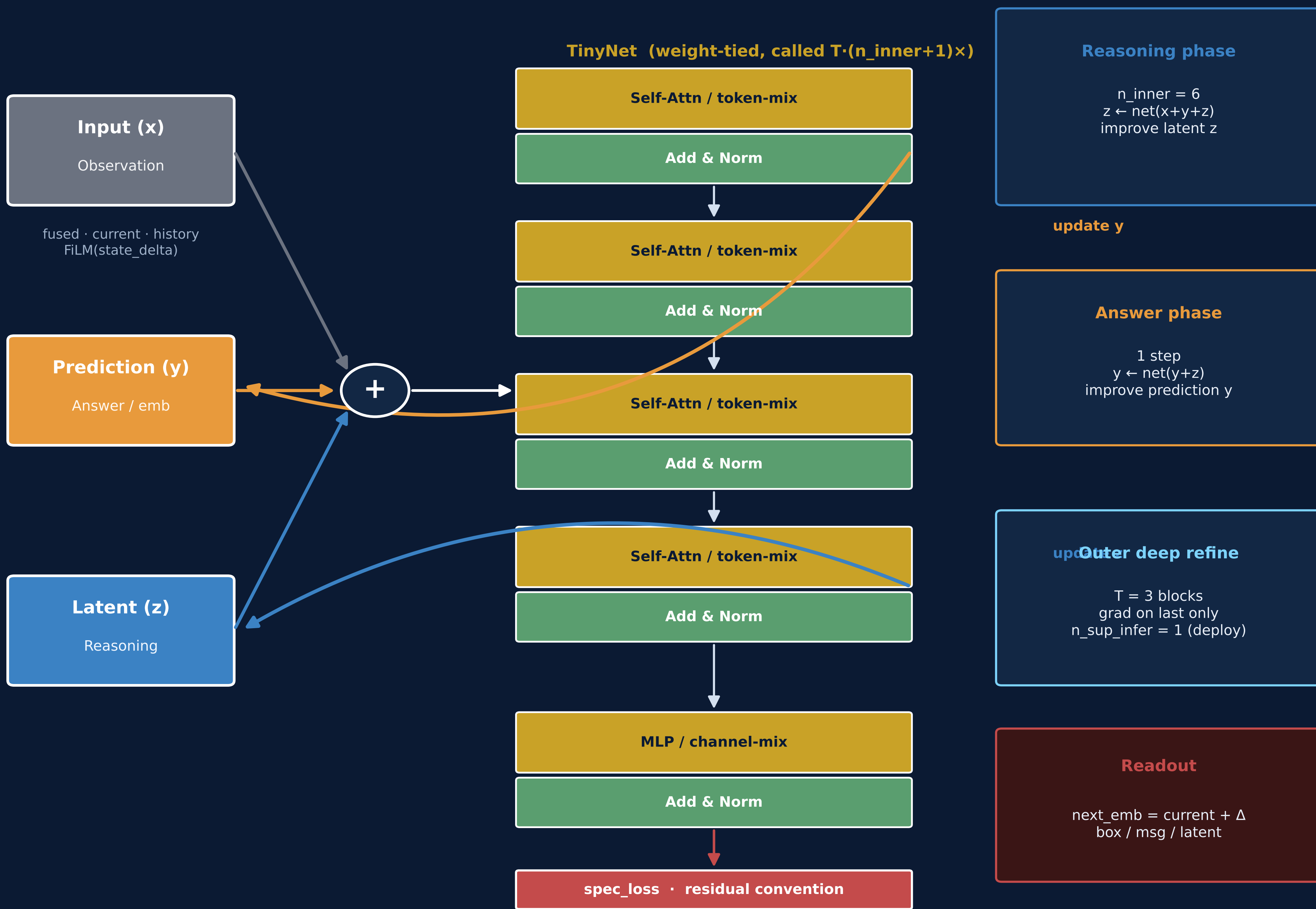
End-to-end dataflow (as-built v9)

Dims are the live checkpoint contract — see DESIGN.md §v9



RecursiveTRM — model pattern

Weight-tied TinyNet · two latents (y prediction, z reasoning) · FiLM observation x



Module patterns & parameter ledger

Module	Params	Pattern
EvidenceEncoder	0.12M	frame|objs|text|action → fused[32,5]; conf×freshness fade
HRMBackbone	2.11M	slow (real) + fast (30 Hz) two-rate state; gain_head zero-init
RecursiveTRM	9.97M	FiLM observe → (y,z) recursion → residual next_emb[512]
TQSA	0.13M	text queries 4×4 spatial grid → token[128] + heat
RelationalHead	2.36M	12 queries × objs/text/latent/action — 97% plan ablation
ChronoQueryPlanner	1.90M	5 time-queries × memory → plan[5,7] + grip + waypoints
InnovationCorrector	0	EMA innovation → trust τ; delta brake / absolute hold
YOLO-World-S	~13M*	frozen vision+text; *not in trainable budget

Hard caps: fusion≤5M · drift≤1.5M · planner≤2.5M · trainable < 9M heads + TRM slot ~10M

JEPA runtime pattern — real vs dream

Core claim: dream evidence is faded evidence on the SAME path as train modality_dropout

design: 1 real + 14 dream @ 30 Hz · eval often uses perception_period=2

REAL tick

1. YOLO perceive → standardized embs
2. EvidenceEncoder (full conf)
3. HRM slow-step + fast
4. TRM forward_full → next_emb
5. Relational + Planner → plan
6. Corrector: innovate, update τ
7. Execute plan row 0; hold boxes

DREAM tick

1. frame token = corrected next_emb
2. boxes HELD from last real
3. weight *= staleness_decay^k (0.9^k)
4. SAME EvidenceEncoder path
5. HRM fast only (slow held)
6. TRM → planner → trust-scaled plan
7. feed corrected latent forward

Evidence fade identity

train: modality_dropout fades box/geometry weights by U[0,1)
infer dream: confidence × 0.9^k · missed detect → weight 0 · last-action token never faded

Special loss functions

TRM spec_loss

$L = 1.0 \cdot (1 - \cos(\hat{y}, y)) + 0.5 \cdot \mathrm{MSE}(\hat{y}, y)$
on CANONICAL standardized embeddings — no LayerNorm inside the loss
deep supervision via refine_forward · n_sup=3 train / n_sup_infer=1 deploy

Stage A world-model

fusion + HRM + TRM · deployment-exact 15-tick rollouts between real frames
spec_loss on predicted next frame emb (and box when enabled)

Stage B planner BC

split_planner_loss: MSE on pose dims + BCE on gripper logit
+ smoothness (2nd-diff) · optional row0_weight · waypoints MSE masked
centering_loss / depth_loss: IBVS-shaped aux for last-cm xy / z

Optional auxiliaries

modality_consistency_loss: MSE(fused_dropped, fused_full.detach())
aligns train dropout with dream fade · waypoint_loss on metric EEF disp

What is NOT a learned loss

PhasedIBVS / grasp-offset / place_at are calibrated constants (0 params)
InnovationCorrector trust is EMA ratio — not backpropagated

Tasks, training stages, evaluation tracks

LIBERO-Object suite (primary)

T0
alphabet soup → basket

assisted 0.75

T1
cream cheese → basket

assisted 0.30

T2
salad dressing → basket

perception hard

Training

A train_full stage A — WM (fusion/HRM/TRM)
B stage B — freeze WM, BC planner
   teacher_bc — distill PhasedIBVS successes
data LIBERO demos → .npz shards (BudgetGuard)
device prefer MPS / CUDA; tests CPU-mock only

Evaluation tracks

unaided   plan only → currently 0.0 success
assisted plan + PhasedIBVS constants
wind tunnel eval.bench --synthetic (no sim)
mock-env always required (no LIBERO deps)
honesty   assisted ≠ policy competence

Plan / action contract

plan shape [5, 7] — rows = timesteps, cols = servos (EEF Δxyz + rot + grip)
action_space="delta" (LIBERO): low trust BRAKES → 0 · "absolute" (Pi PWM): HOLD-blend
canonical emb space everywhere; TRM residual: next = current + Δ

Design claims (what is / is not novel)

Distinctive

- Detector-as-text-encoder (no separate LM)
- Dream evidence ≡ faded evidence (one code path)
- Test-enforced parameter + disk budgets
- Forensic defect taxonomy + lever-arm handoff

Composition of known ideas

- JEPA-style predictive world models
- Recursive / Tiny recursive transformers (TRM)
- Cross-attention planners / relational tokens
- Classical IBVS / hand-eye calibration

Explicit non-claims

- No unaided LIBERO success claimed (still 0)
- Assisted numbers ≠ learned policy competence
- Sim-only; Pi 5 is design target, not reported deploy
- Not competing with OpenVLA-scale absolute SR

Binding docs: DESIGN.md · microvla/config.py · TRM.py · paper/MANUSCRIPT.md

Generate: .venv/bin/python paper/render_architecture_pdf.py